Coding theory:

Conding theory plays an important role in Communication theory.

The Communication process takes place in a Variety of ways like telephone, Sending telegram etc. This process Consists of a Sender (or) Source, Storage medium and a receiver.

Sender or Source → Channel Storage medium → receiver

The storage medium transmits the message from sender to receiver.

It may be a telegraphic exchange or a number of computers connected by communication lines.

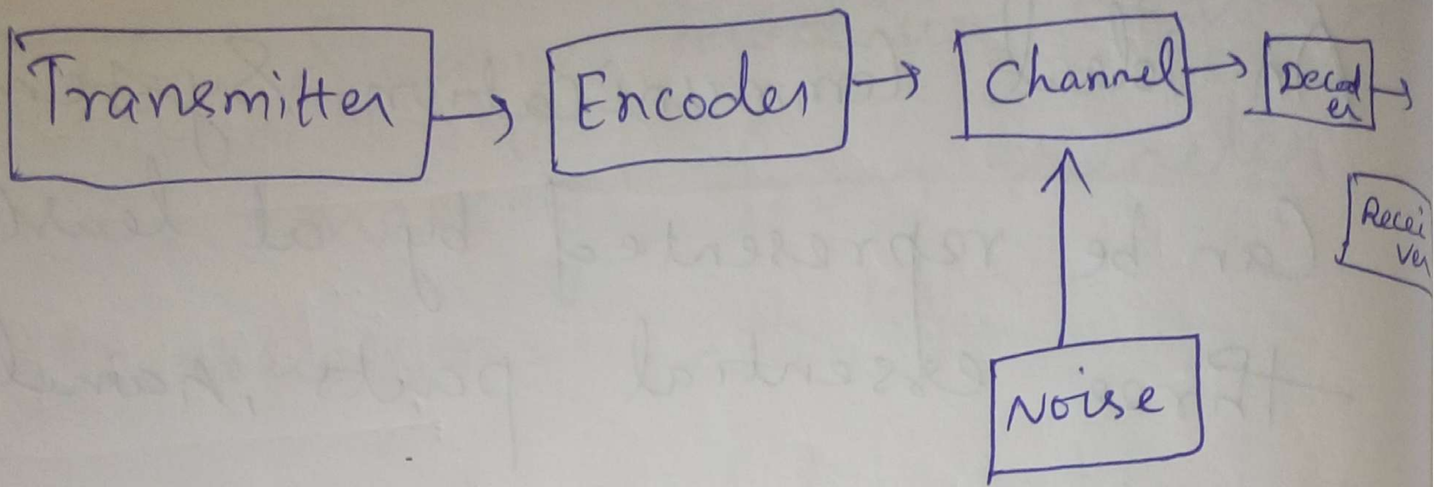
But, this communication channel is subjected to a number of disturbances, distorting the message. Any such disturbance is called a noise. For example, in a telephone conversion the noise may be due to a passing vehicle or other voices, etc.

An ideal Communication System Can be represented by at least three essential parts, namely

1. Transmitter, Sender or Source
2. Channel or Storage medium
3. Receiver.

A device that can be used to improve the efficiency of the Communication Channel is an encoder.

Thus, the encoder is used to detect the noises in the transmitted messages.



Along with the encoder, a decoder is also used to transform the encoded message into an acceptable form.

When message are expressed in some language, they are transformed into another language understood by the transmitter and the receiver. Such a message is called a coded message.

The process of transformation is called encoding process. The inverse process is called as the decoding process.

In our discussion, we are interested in the binary language for encoding and decoding process, ie we will deal with only a binary channel in which the encoder will transform an input message into a binary string consisting of the symbols 0 and 1. Decoding is only the inverse operation of encoding.

$B^n = \underbrace{B \times B \times \cdots \times B}_{n \text{ times}}$

Group Code:

B^n is written as $(x_1, x_2 \cdots x_n)$ or simply $x_1 x_2 \cdots x_n$. Note that B^n has 2^n elements

Definite:

If $B = (0,1)$ then

$$B^n = \{x_1, x_2, \cdots x_n / x_i \in B, i = 1, 2, 3, \cdots n\}$$

is a group under the binary operation of addition modulo 2, denoted by $\oplus$. This group $(B^n, \oplus)$ is called a group code.

Let us now prove that $(B^n, \oplus)$ is a group.

If $(x_1, x_2 \ldots x_n)$ and $(y_1, y_2 \ldots y_n) \in B^n$

then $(x_1, x_2 \ldots x_n \oplus y_1, y_2 \ldots y_n) =$

$(x_1 +_2 y_1, x_2 +_2 y_2, \ldots x_n +_2 y_n) \in B^n$

$\oplus$	0	1
0	0	1
1	1	0

Since $x_i +_2 y_i = 1$ or 0 as $0 +_2 0 = 0$,

$0 +_2 1 = 1$

$1 +_2 0 = 1$

$1 +_2 1 = 0$.

Note! The operation $+_2$ is also called binary addition.

$(0, 0, \ldots 0)$ is the identity element of B^n.

Also the inverse of $x_1, x_2 \ldots x_n$ is itself.

(Ex:) if $x = 110001 \in B^6$ then

$x \oplus x = 110001 \oplus 110001$

$= 000000 \in B^6$.

Moreover, the order of the group $(B^n, \oplus)$ is 2^n.

Hence $(B^n, \oplus)$ is a group - It is abelian.

In general, any code which is a group under the operation $\oplus$ is called a group code.

Hamming Codes:

It was developed by Richard Hamming in 1950.

The codes obtained by introducing additional digits called parity digits to the digits in the original message are called Hamming codes.

If the original message is a binary string of length m, the Hamming encoded message is string of length n, $(n > m)$.

→ **weight of a binary string:**

Let $x \in B^n$ then weight of x is the number of 1's in x and it is denoted by $|x|$.

Example:

$$|101011| = 4$$

of the n digits, m digits are used to represent the information part of the message and the remaining $(n-m)$ digits are used for the detection and correction of errors in the message received.

In Hamming's single-error detecting code of length n, the first $(n-1)$ digits contain the information part of the message and the last digit is made either 0 or 1.

If the digit introduced in the last position gives an even number/ odd number of 1's in the encoded word of length n, the extra digit is called an even/odd parity check.

We define a one-to-one function $e: B^m \longrightarrow B^n$. The function e is called an (m, n) encoding function. If $x = (x_1, x_2 \cdots x_n) \in B^m$, then $e(x)$ is called the code word representing x.

The encoding function

$e : B^m \to B^{m+1}$ is called the

even parity $(m, m+1)$ check code

if $x = x_1 x_2 \ldots x_m \in B^m$, define

$$e(x) = x_1 x_2 \ldots x_m x_{m+1}$$

where $x_{m+1} = \begin{cases} 0, & \text{if } |x| \text{ is even} \\ 1, & \text{if } |x| \text{ is odd} \end{cases}$

and $|x|$ is the number of 1's present in x, called the weight of x.

It may be noted that x_{m+1} is zero if and only if the number of 1's in x is even number.

Thus, every code word $c(x)$ has even weight.

Hamming distance:

Let $x, y \in B^m$ then the hamming distance $H(x, y)$ is the weight of $x \oplus y$.

Ex: Let $x = 1011$ & $y = 0101$

then $H(x, y) = |1011 \oplus 0101| = |1110| = 3$

Note: The minimum hamming distance between of all pair of encoded word is the minimum distance.

① Find the minimum distance of a code $x = 10110$, $y = 11110$ & $z = 10011$.

Sol:

$$H(x, y) = |x \oplus y| = |10110 \oplus 11110|$$

$$= |01000| = 1$$

$$H(y, z) = |y \oplus z| = |01101| = 3$$

$$H(z, x) = |z \oplus x| = |00101| = 2$$

$\therefore$ minimum distance is 1.

Note: The term 'code' used is sometime called an (m, n)

encoding function, which is a one-to-one function $e: B^m \to B^n$ (where $n > m$). If $b \in B^m$ is the original word, then $e(b)$ is the code word or encoded word representing b.

Definition:

An (m, n) encoding function $e: B^m \to B^n$ $(n > m)$ is called a Group Code if $e(B^m)$ is a subgroup of B^n.

(or)

An (m, n) encoding function $e: B^m \to B^n$ is called a group code if the range of e ie $e(B^m) = \{e(x) : x \in B^m\}$ is a subgroup of B^n.

Ex:

Consider (3, 6) encoding function $e : B^3 \to B^6$ defined by

$x \in B^3$	Code word $e(x) \in B^6$
000	000000
001	001100
010	010011
011	011111
100	100101
101	101001
110	110110
111	111010

Show that this encoding function is a group code.

Sol:

Let $S = \{000000, 001100, 010011,$
$011111, 100101, 101001, 110110, 111010\}$

Lete first show that S is a subgroup of B^6. Here the identity element $000000 \in B^6$.

Each element is its own inverse.

It can be verified that, if $x, y \in S$, then $x \oplus y \in S$.

Thus, S is a subgroup of B^6.

Hence, the given encoding function is a group code.

(1) Find the code words generated by the encoding function $e : B^2 \longrightarrow B^5$ w.r.t the parity check matrix.

$$H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol:

Let $e : B^m \longrightarrow B^n$ where $m < n$ be the encoding function.

i) $G = [I_m / A]$ be the generated matrix

ii) $H = [\underbrace{A^T}_{(n-m)\times m \text{ matrix}} / I_{n-m}]$ be the parity check matrix.

$$H = \left[\begin{array}{cc|ccc} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}\right] = \left[A^T / I_{n-m}\right]$$

where $n = 5$ & $m = 2$

Generator matrix : G is

$$G = [\underbrace{I_m}_{m\times m \text{ unit matrix}} / \underbrace{A}_{m\times(n-m) \text{ matrix to be chosen suitably}}] = \left[\begin{array}{cc|ccc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array}\right]$$

Now $B^2 = \{00, 01, 10, 11\}$ & $e(w) = wG$

$$e(00) = [0\ 0]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [00000]$$

$$e(01) = [0\ 1]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [01011]$$

$$e(10) = [1\ 0]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [10011]$$

$$e(11) = [1\ 1]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [11000]$$

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∴ The code words generated by H are 00000, 01011, 10011 & 11000.

② Find the code words generated by the parity check matrix.

$$H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

When the encoding function is $e : B^3 \to B^6$.

Where $B^3 = [000, 001, 010, 100, 011, 101, 110, 111]$

The code words are 000000, 001011, 010101, 100111, 011110, 101100, 110010 & 111001.

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Theorem:

A Code can detect atmost k errors iff the minimum distance between any two code words is atleast (k+1).

Theorem:

A Code can correct a set of atmost k errors iff the minimum distance between any two code word is atleast (2k+1).

① Consider the following (2,5) encoding functione
3

s.t.

$e(00) = 00000$

$e(01) = 01110$

$e(10) = 10111$

$e(11) = 11111$

} Code words

i) How many errors will e detect?

ii) How many errors will e correct?

② For the following encoding function, find the minimum distance between the code words. State also the error-detecting & error correcting

Capabilities of each code.

$e(00) = 0000000000$

$e(10) = 1111100000$

$e(01) = 0000011111$

$e(11) = 1111111111$

Sol:

Let $x_1 = 0000000000$

$x_2 = 1111100000$

$x_3 = 0000011111$

$x_4 = 1111111111$

then

$|x_1 \oplus x_2| = H(x_1, x_2) = |1111100000| = 5$

$|x_1 \oplus x_3| = H(x_1, x_3) = |0000011111| = 5$

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$$|x_1 \oplus x_4| = H(x_1, x_4) = |1111111111| = 10$$

$$|x_2 \oplus x_3| = |1111111111| = 10$$

$$|x_2 \oplus x_4| = |0000011111| = 5$$

$$|x_3 \oplus x_4| = |1111100000| = 5$$

minimum distance of the code word e is 5.

W.K.T. a code will detect atmost K errors if the minimum distance between any two code words is atleast K+1.

minimum distane $= k+1 = 4+1 = 5$

∴ The Code will detect atmost 4 errors.

Also w.k.T. a code will Correct atmost k errors if the minimum distance between any two Code words is atleast $2k+1$.

∴ The minimum distance $= 5 = 2k+1$

$= 2(2)+1$

∴ The Code will Correct atmost 2 errors.